PROBLEM 33

Two Repelling Particles on a Rotating Parabola

Original statement · Problems.pdf, p. 21

Rotating Wire + Two Particles

Two heavy point particles, P_1 and P_2, of mass m each, slide without friction along a rigid wire parabola given by the equation $z = x^2/(2L)$, where L is a known length. The wire rotates about its axis with a fixed, known angular velocity ω. The two particles repel each other electrically with a force of magnitude $F_E = k/d^2$, where k is a known constant and d is the distance between the particles. We define the rotating reference frame $Oxyz$ attached to the parabola as shown in the figure below.

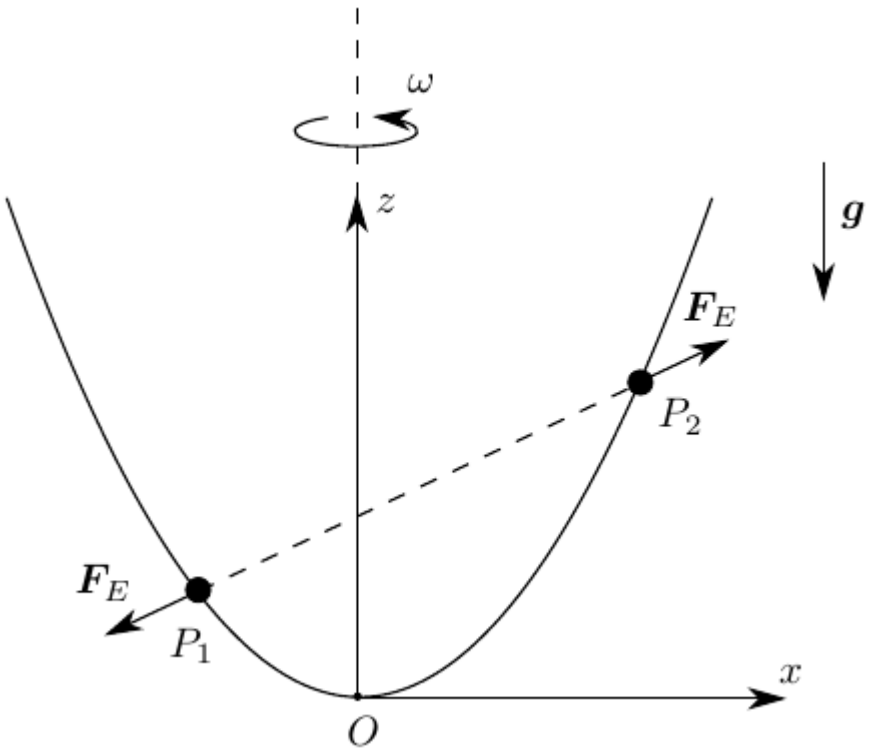


Figure: Parabola z = x²/(2L) in the plane Oxz rotating with angular velocity ω about Oz, particles P1 (left branch) and P2 (right branch) with the repulsive forces F_E along the line P1P2, gravity g downward (Problems.pdf, p. 21).

(a) Write down the position, velocity, and acceleration vectors of each particle relative to $Oxyz$ in a generic instant of time, as a function of x_1, x_2 (the position of P_1 and P_2 along the Ox axis) and their derivatives.

(b) Write down Newton's second law for each particle in the $Oxyz$ reference frame.

(c) If the rotation of the wire is maintained by a motor, how much torque must it produce in the Oz direction to keep ω constant as a function of the variables of the problem and their derivatives?

(d) Find out the equilibrium configurations of the system in the reference frame $Oxyz$ as a function of the parameters $\gamma = g/(\omega^2 L)$ and $\kappa = k/(m\omega^2 L^3)$.

Consider now that P_1 and P_2 are constrained to move symmetrically on each branch of the parabola, and that $\gamma = \kappa = 2$. Initially, $x_1(0) = L$ and $v_1(0) = L\omega$ (with $\dot{x}_1(0) > 0$).

(e) Write down the equation that gives the minimum and maximum values of x_1 occurring in this case. *Hint: this equation is a third-order polynomial in x_1.*

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** two particles P_1, P_2 of mass m slide on a smooth parabolic wire $z = x^2/(2L)$ that spins about its vertical axis Oz at constant ω. They repel each other with $F_E = k/d^2$ along the line that joins them, and both are heavy.
- **Why the rotating frame $Oxyz$.** In $Oxyz$ the wire is fixed, so each particle is located by a single coordinate x_i and its relative kinematics is that of a point on a fixed plane curve. The frame rotates uniformly about a fixed axis through O, so the inertia forces reduce to

the centrifugal and Coriolis forces (Slide 28), and the centrifugal force has a potential (Slide 30).

- **Hypotheses:**
  1. The frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with O the vertex of the parabola and $\mathbf{k}_0$ vertical upward, is inertial.
  2. The wire is smooth and bilateral (each particle is a bead): its reaction on P_i is perpendicular to the wire, with components in the plane of the wire and along $\mathbf{j}_1$.
  3. ω is kept constant by the motor; the axis bearings exert no torque about Oz.
  4. Electric repulsion is a central internal force: equal and opposite on the two particles, along $\overrightarrow{P_2P_1}$.
- **Degrees of freedom:** two, the abscissas x_1 and x_2 in $Oxyz$. In part (e) the symmetry constraint $x_2 = -x_1$ leaves one.

Symbol	Meaning	SI unit
m	mass of each particle	kg
L	parameter of the parabola $z = x^2/(2L)$ (the radius of curvature at the vertex)	m
ω	constant angular velocity of the wire	rad/s
g	gravitational acceleration	m/s^2
k	electric repulsion constant, $F_E = k/d^2$	N·m^2
$\gamma = g/(\omega^2 L)$	weight versus centrifugal force	1
$\kappa = k/(m\omega^2 L^3)$	repulsion versus centrifugal force	1

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial; $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire, with the wire in the plane Oxz and $\mathbf{k}_1 = \mathbf{k}_0$ along the axis. The two frames share the origin O, which lies on the axis and is fixed.
- **Change of basis** (rotation of angle ωt about $\mathbf{k}_0$, with $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$; columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors and mutually orthogonal, so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega \mathbf{k}_1$; $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because ω is constant.
- **Local geometry of the wire at P_i.** A tangent vector is $\dfrac{d}{dx}\left(x\,\mathbf{i}_1 + \dfrac{x^2}{2L}\mathbf{k}_1\right) = \mathbf{i}_1 + \dfrac{x}{L}\mathbf{k}_1$. With $s_i = \sqrt{1 + x_i^2/L^2}$, the unit tangent and the in-plane unit normal (towards the concave side) are

$$\mathbf{t}_i = \frac{1}{s_i}\left(\mathbf{i}_1 + \frac{x_i}{L}\mathbf{k}_1\right), \qquad \mathbf{n}_i = \frac{1}{s_i}\left(-\frac{x_i}{L}\mathbf{i}_1 + \mathbf{k}_1\right)$$

They are orthogonal, $\mathbf{t}_i \cdot \mathbf{n}_i = \dfrac{1}{s_i^2}\left(-\dfrac{x_i}{L} + \dfrac{x_i}{L}\right) = 0$, and, using $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$,

$$\mathbf{t}_i \times \mathbf{n}_i = \frac{1}{s_i^2}\left[\mathbf{i}_1 \times \mathbf{k}_1 - \frac{x_i^2}{L^2}\,\mathbf{k}_1 \times \mathbf{i}_1\right] = \frac{1}{s_i^2}\left[-\mathbf{j}_1 - \frac{x_i^2}{L^2}\mathbf{j}_1\right] = -\mathbf{j}_1$$

so $\{\mathbf{t}_i, \mathbf{n}_i, -\mathbf{j}_1\}$ is a right-handed orthonormal triad attached to the wire at P_i. The reaction of the smooth wire lies in the plane spanned by $\mathbf{n}_i$ and $\mathbf{j}_1$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative position, velocity and acceleration

Pedagogical Justification: Relative kinematics is computed in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant, so only the scalar components are differentiated. Each $z_i = x_i^2/(2L)$ depends on time through $x_i(t)$, so the chain rule is needed.

For $i = 1, 2$, the position on the wire is

$$\mathbf{r}_1^{P_i} = x_i\,\mathbf{i}_1 + \frac{x_i^2}{2L}\,\mathbf{k}_1$$

Chain rule: $\dfrac{d}{dt}\left(\dfrac{x_i^2}{2L}\right) = \dfrac{2x_i}{2L}\,\dot{x}_i = \dfrac{x_i \dot{x}_i}{L}$. Hence

$$\boxed{\mathbf{v}_1^{P_i} = \dot{x}_i\,\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\,\mathbf{k}_1}$$

Product rule on $x_i\dot{x}_i$: $\dfrac{d}{dt}(x_i\dot{x}_i) = \dot{x}_i\dot{x}_i + x_i\ddot{x}_i$. Hence

$$\boxed{\mathbf{a}_1^{P_i} = \ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\,\mathbf{k}_1}$$

Note $\mathbf{v}_1^{P_i} = s_i\dot{x}_i\,\mathbf{t}_i$, tangent to the wire, as it must be.

(b) Newton's second law in $Oxyz$

> **Pedagogical Justification:** $Oxyz$ is not inertial; Newton's law holds there if the inertia forces are added, $m\mathbf{a}_1^{P_i} = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). Projecting on the tangent $\mathbf{t}_i$ eliminates the unknown reaction and gives the equation of motion; the other two projections give the reaction.

Actual forces on P_i:

- Weight: $-mg\,\mathbf{k}_1$.
- Reaction of the smooth wire: $\mathbf{N}_i = N_{in}\,\mathbf{n}_i + N_{iy}\,\mathbf{j}_1$ (so $\mathbf{N}_i\cdot\mathbf{t}_i = 0$).
- Electric repulsion. With $\Delta x = x_1 - x_2$ and $\Delta z = \dfrac{x_1^2 - x_2^2}{2L} = \Delta x\,\dfrac{x_1 + x_2}{2L}$,

$$\overrightarrow{P_2P_1} = \Delta x\left[\mathbf{i}_1 + \frac{x_1+x_2}{2L}\mathbf{k}_1\right], \qquad d^2 = |\overrightarrow{P_2P_1}|^2 = \Delta x^2\left[1 + \frac{(x_1+x_2)^2}{4L^2}\right]$$

The force on P_1 has magnitude k/d^2 and the direction of $\overrightarrow{P_2P_1}/d$, and the force on P_2 is opposite:

$$\mathbf{F}_{E1} = \frac{k}{d^3}\,\Delta x\left[\mathbf{i}_1 + \frac{x_1+x_2}{2L}\mathbf{k}_1\right] = -\mathbf{F}_{E2}$$

Inertia forces ($\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so $\mathbf{F}_A = \mathbf{F}_E^{\text{Euler}} = \mathbf{0}$):

- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \omega \mathbf{k}_1 \times \left(x_i \mathbf{i}_1 + \frac{x_i^2}{2L} \mathbf{k}_1 \right) = \omega x_i \, \mathbf{j}_1$, then $\boldsymbol{\omega}_{10} \times \omega x_i \mathbf{j}_1 = \omega^2 x_i \left(\mathbf{k}_1 \times \mathbf{j}_1 \right) = -\omega^2 x_i \, \mathbf{i}_1$, so $\mathbf{F}_{C,i} = m\omega^2 x_i \, \mathbf{i}_1$ (outward from the axis, Slide 30).
- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i} = \omega \mathbf{k}_1 \times \left(\dot{x}_i \mathbf{i}_1 + \frac{x_i \dot{x}_i}{L} \mathbf{k}_1 \right) = \omega \dot{x}_i \, \mathbf{j}_1$, so $\mathbf{F}_{COR,i} = -2m\omega \dot{x}_i \, \mathbf{j}_1$.

Newton's second law for each particle (vector form):

$$m \left[\ddot{x}_i \, \mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i \ddot{x}_i}{L} \, \mathbf{k}_1 \right] = -mg \, \mathbf{k}_1 + N_{in} \, \mathbf{n}_i + N_{iy} \, \mathbf{j}_1 + \mathbf{F}_{Ei} + m\omega^2 x_i \, \mathbf{i}_1 - 2m\omega \dot{x}_i \, \mathbf{j}_1, \qquad i = 1, 2$$

Projection on $s_i \mathbf{t}_i = \mathbf{i}_1 + \frac{x_i}{L} \mathbf{k}_1$ (no reaction). Left side: $m \left[\ddot{x}_i + \frac{x_i}{L} \cdot \frac{\dot{x}_i^2 + x_i \ddot{x}_i}{L} \right] = m \left[\left(1 + \frac{x_i^2}{L^2} \right) \ddot{x}_i + \frac{x_i \dot{x}_i^2}{L^2} \right]$. Right side: weight $-mg \, x_i / L$; centrifugal $m\omega^2 x_i$; Coriolis 0; electric, for P_1,

$$\mathbf{F}_{E1} \cdot \left(\mathbf{i}_1 + \frac{x_1}{L} \mathbf{k}_1 \right) = \frac{k}{d^3} \left(x_1 - x_2 \right) \left[1 + \frac{x_1 (x_1 + x_2)}{2L^2} \right]$$

and for P_2 the same with the indices exchanged (because $\mathbf{F}_{E2} = -\mathbf{F}_{E1}$ and $-\Delta x = x_2 - x_1$). The **equations of motion** are

$$\boxed{m \left[\left(1 + \frac{x_1^2}{L^2} \right) \ddot{x}_1 + \frac{x_1 \dot{x}_1^2}{L^2} \right] = m \left(\omega^2 - \frac{g}{L} \right) x_1 + \frac{k}{d^3} \left(x_1 - x_2 \right) \left[1 + \frac{x_1 (x_1 + x_2)}{2L^2} \right]}$$

$$\boxed{m \left[\left(1 + \frac{x_2^2}{L^2} \right) \ddot{x}_2 + \frac{x_2 \dot{x}_2^2}{L^2} \right] = m \left(\omega^2 - \frac{g}{L} \right) x_2 + \frac{k}{d^3} \left(x_2 - x_1 \right) \left[1 + \frac{x_2 (x_1 + x_2)}{2L^2} \right]}$$

with $d^3 = |x_1 - x_2|^3 \left[1 + \frac{(x_1 + x_2)^2}{4L^2} \right]^{3/2}$.

Projection on $\mathbf{j}_1$: $0 = N_{iy} - 2m\omega \dot{x}_i$, so $\boxed{N_{iy} = 2m\omega \dot{x}_i}$: the wire pushes each bead sideways to cancel the Coriolis force.

Projection on $s_i\mathbf{n}_i = -\dfrac{x_i}{L}\mathbf{i}_1 + \mathbf{k}_1$. Left side: $m\left[-\dfrac{x_i\ddot{x}_i}{L} + \dfrac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\right] = \dfrac{m\dot{x}_i^2}{L}$. Right side: $N_{in}s_i - mg - \dfrac{m\omega^2 x_i^2}{L} + \mathbf{F}_{Ei}\cdot(s_i\mathbf{n}_i)$, where for P_1

$$\mathbf{F}_{E1}\cdot\left(-\frac{x_1}{L}\mathbf{i}_1 + \mathbf{k}_1\right) = \frac{k\,\Delta x}{d^3}\left[-\frac{x_1}{L} + \frac{x_1 + x_2}{2L}\right] = \frac{k\,\Delta x}{d^3}\cdot\frac{x_2 - x_1}{2L} = -\frac{k\,\Delta x^2}{2Ld^3}$$

and the same value is found for P_2. Therefore

$$\boxed{N_{in} = \frac{m}{s_i}\left[\frac{\dot{x}_i^2}{L} + g + \frac{\omega^2 x_i^2}{L}\right] + \frac{k\,(x_1 - x_2)^2}{2L\,s_i\,d^3}}, \qquad s_i = \sqrt{1 + \frac{x_i^2}{L^2}}$$

(c) Motor torque that keeps ω constant

> **Pedagogical Justification:** The motor torque acts on the wire; the wire acts on the particles through the reactions $\mathbf{N}_i$, and the particles act back with $-\mathbf{N}_i$. The wire rotates at constant ω about a fixed axis and is symmetric about Oz (its products of inertia I_{xz}, I_{yz} vanish), so its own angular momentum $I_{zz}\omega\,\mathbf{k}_1$ is constant, and the moment about Oz of all actions on it must vanish.

Balance of moments about Oz on the wire: $M + \sum_i \left[\mathbf{r}^{P_i}\times(-\mathbf{N}_i)\right]\cdot\mathbf{k}_1 = 0$. Expand the cross product with $\mathbf{r}^{P_i} = x_i\mathbf{i}_1 + z_i\mathbf{k}_1$ and $\mathbf{N}_i = N_{ix}\mathbf{i}_1 + N_{iy}\mathbf{j}_1 + N_{iz}\mathbf{k}_1$; its $\mathbf{k}_1$ component is $x_iN_{iy} - 0\cdot N_{ix} = x_iN_{iy}$. With $N_{iy} = 2m\omega\dot{x}_i$ from part (b):

$$\boxed{M = x_1N_{1y} + x_2N_{2y} = 2m\omega\,(x_1\dot{x}_1 + x_2\dot{x}_2)}$$

Check with the angular momentum of the particles in S_0. By Notes Eq. 3.26, $\mathbf{v}_0^{P_i} = \mathbf{v}_1^{P_i} + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^{P_i} = \dot{x}_i\mathbf{i}_1 + \omega x_i\mathbf{j}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1$, whose $\mathbf{k}_1$-moment about O is $m\left[x_i(\omega x_i) - 0\cdot\dot{x}_i\right] = m\omega x_i^2$. So $H_{Oz} = m\omega(x_1^2 + x_2^2)$ and, by the product and chain rules, $\dot{H}_{Oz} = 2m\omega(x_1\dot{x}_1 + x_2\dot{x}_2)$. The weights are parallel to Oz and the electric pair is central ($(\mathbf{r}^{P_1} - \mathbf{r}^{P_2})\times\mathbf{F}_{E1} = \mathbf{0}$), so only the reactions have a moment about Oz, and that moment equals M. Both routes agree with the official key (Problems.pdf, p. 75).

(d) Equilibrium configurations in $Oxyz$

> **Pedagogical Justification:** A relative equilibrium has $\dot{x}_i = \ddot{x}_i = 0$; then the two equations of motion become algebraic. Writing them with $u_i = x_i/L$ and dividing by $m\omega^2 L$ shows that only γ and κ matter.

With $\dot{x}_i = \ddot{x}_i = 0$, $x_i = Lu_i$ and $d = LD$, $D = |u_1 - u_2|\sqrt{1 + (u_1+u_2)^2/4}$, dividing by $m\omega^2 L$:

$$f_1 \equiv (1-\gamma)\,u_1 + \frac{\kappa}{D^3}(u_1 - u_2)\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0, \qquad f_2 \equiv (1-\gamma)\,u_2 + \frac{\kappa}{D^3}(u_2 - u_1)\left[1 + \frac{u_2(u_1+u_2)}{2}\right] = 0$$

Sum. With $S = u_1 + u_2$, the electric terms add to $\dfrac{\kappa}{D^3}(u_1 - u_2)\left[\dfrac{u_1 S}{2} - \dfrac{u_2 S}{2}\right] = \dfrac{\kappa\,(u_1-u_2)^2 S}{2D^3}$, so

$$f_1 + f_2 = S\left[(1-\gamma) + \frac{\kappa\,(u_1-u_2)^2}{2D^3}\right] = 0$$

Either $S = 0$ (symmetric configuration) or $\gamma - 1 = \dfrac{\kappa(u_1-u_2)^2}{2D^3} > 0$.

Case $\gamma < 1$. If $S = 0$, $u_2 = -u_1$, $D = 2|u_1|$ and $f_1 = u_1\left[(1-\gamma) + \dfrac{2\kappa}{8|u_1|^3}\right]$: both terms of the bracket are positive and $u_1 \neq 0$ (the particles cannot coincide), so $f_1 \neq 0$. If $S \neq 0$ the bracket of the sum is positive. **No equilibrium exists.** (Centrifugal force beats gravity everywhere and the repulsion only pushes the particles further out.)

Case $\gamma = 1$. Both alternatives require $\kappa = 0$. For $\kappa > 0$ there is no equilibrium; only without repulsion ($\kappa = 0$) is every configuration an equilibrium (gravity and centrifugal force balance along the wire at every point: $(\omega^2 - g/L)x_i = 0$).

Case $\gamma > 1$, symmetric ($x_2 = -x_1$). $f_1 = u_1\left[(1-\gamma) + \dfrac{\kappa}{4|u_1|^3}\right] = 0$ gives $|u_1|^3 = \dfrac{\kappa}{4(\gamma-1)}$:

$$\boxed{x_1 = -x_2, \qquad x_1^2 = \left[\frac{\kappa L^3}{4(\gamma-1)}\right]^{2/3}}$$

Case $\gamma > 1$, asymmetric ($S \neq 0$). Insert $\dfrac{\kappa}{D^3} = \dfrac{2(\gamma-1)}{(u_1-u_2)^2}$ from the sum into $f_1 = 0$ and divide by $(\gamma - 1)$:

$$-u_1 + \frac{2}{u_1-u_2}\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0 \implies u_1(u_1-u_2) = 2 + u_1(u_1+u_2) \implies \boxed{x_1 x_2 = -L^2}$$

The particles are on opposite branches and the chord P_1P_2 passes through the focus $(0, L/2)$ of the parabola. Then $u_2 = -1/u_1$, $S = u_1 - 1/u_1$, $u_1 - u_2 = u_1 + 1/u_1$, and $1 + S^2/4 = (u_1 + 1/u_1)^2/4$, so $D^3 = |u_1 + 1/u_1|^6/8$ and the sum condition becomes

$$\gamma - 1 = \frac{4\kappa}{(u_1 + 1/u_1)^4} \implies \left|u_1 + \frac{1}{u_1}\right| = \left(\frac{4\kappa}{\gamma-1}\right)^{1/4}$$

Since $|u_1 + 1/u_1| \geq 2$, with equality only for $|u_1| = 1$ (which is the symmetric equilibrium), **asymmetric equilibria exist if and only if** $\kappa > 4(\gamma - 1)$, i.e. exactly when the symmetric equilibrium lies at $|x_1| > L$. They come in mirror pairs, $u_1 = \frac{1}{2}\left[q \pm \sqrt{q^2 - 4}\right]$ with $q = (4\kappa/(\gamma-1))^{1/4}$ and $u_2 = -1/u_1$ (plus the configurations with the labels exchanged). No equilibrium has both particles on the same branch.

(e) Symmetric motion with $\gamma = \kappa = 2$: turning points

> **Pedagogical Justification:** In $Oxyz$ the wire (and the symmetry constraint) is fixed, $\boldsymbol{\omega}_{10}$ is constant and the origin is fixed. The reactions and the Coriolis forces do no work in S_1 (Slide 31), and the weights, the electric pair and the centrifugal forces have potentials (Slides 29–30). The mechanical energy of the pair in S_1 is therefore conserved, and its turning points ($\dot{x}_1 = 0$) give the extreme values of x_1.

Put $x_1 = x > 0$, $x_2 = -x$ (the repulsion becomes infinite at $x = 0$, so x_1 stays positive). Then $d = 2x$ and, for each particle, $|\mathbf{v}_1^{P_i}|^2 = \dot{x}^2\left(1 + x^2/L^2\right)$.

- Kinetic energy: $T_1 = 2 \cdot \frac{1}{2} m\dot{x}^2\left(1 + \dfrac{x^2}{L^2}\right) = m\dot{x}^2\left(1 + \dfrac{x^2}{L^2}\right)$.
- Gravity: $2\,mg\,\dfrac{x^2}{2L} = \dfrac{mgx^2}{L}$. Centrifugal: $2\left(-\frac{1}{2}m\omega^2 x^2\right) = -m\omega^2 x^2$. Electric: $\dfrac{k}{d} = \dfrac{k}{2x}$ (its gradient gives the repulsion k/d^2).

$$E_1 = m\dot{x}^2\left(1+\frac{x^2}{L^2}\right) + m\left(\frac{g}{L}-\omega^2\right)x^2 + \frac{k}{2x} = \text{const}$$

Initial value. $|\mathbf{v}_1^{P_1}(0)| = L\omega$ at $x = L$ gives $2\dot{x}(0)^2 = L^2\omega^2$, so $T_1(0) = mL^2\omega^2$ and $E_1 = mL^2\omega^2 + m(g/L-\omega^2)L^2 + k/(2L)$. Dividing by $m\omega^2L^2$, with $u = x/L$:

$$\frac{E_1}{m\omega^2L^2} = \gamma + \frac{\kappa}{2} = 3 \qquad (\gamma = \kappa = 2)$$

Turning points ($\dot{x} = 0$): $(\gamma-1)u^2 + \dfrac{\kappa}{2u} = \gamma + \dfrac{\kappa}{2}$, i.e. $u^2 + \dfrac{1}{u} = 3$. Multiply by $u > 0$:

$$\boxed{\frac{x_1^3}{L^3} - 3\,\frac{x_1}{L} + 1 = 0}$$

Roots. With $u = 2\cos\theta$ and $4\cos^3\theta - 3\cos\theta = \cos 3\theta$: $8\cos^3\theta - 6\cos\theta + 1 = 2\cos 3\theta + 1 = 0$, so $\cos 3\theta = -\frac{1}{2}$ and $\theta = 2\pi/9,\ 4\pi/9,\ 8\pi/9$. The roots are $u = 2\cos(2\pi/9) = 1.532$, $2\cos(4\pi/9) = 0.347$ and $2\cos(8\pi/9) = -1.879$ (rejected, $x_1 > 0$). Since the starting point $u = 1$ lies between the two positive roots (there $u^2 + 1/u = 2 < 3$),

$$x_{1,\min} = 2L\cos\frac{4\pi}{9} \approx 0.347\,L, \qquad x_{1,\max} = 2L\cos\frac{2\pi}{9} \approx 1.532\,L$$

and x_1 oscillates between them. The key also lists $x_1^3/L^3 - 3x_1/L - 1 = 0$ for $x_1 < 0$: that is the mirror image (a start on the negative branch) and does not apply to $x_1(0) = L > 0$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Effective spring along the wire.** Gravity and the centrifugal force combine into $m(\omega^2 - g/L)x_i$ in the tangential equation: a restoring force if $\gamma > 1$ (gravity wins and pulls the beads to the vertex) and a repelling one if $\gamma < 1$. Equilibrium requires the repulsion to be balanced by this restoring force, hence the condition $\gamma > 1$.

- **Limits.** $\kappa \to 0$: the symmetric equilibrium collapses to $x_1 = x_2 = 0$ (no repulsion to separate the beads). $\gamma \to 1^+$: $x_1^2 \to \infty$, the restoring force vanishes and the beads are pushed to infinity.
- **Torque sign.** $M = \frac{d}{dt}\left[m\omega(x_1^2 + x_2^2)\right]$: when the beads move outward the moment of inertia of the system grows and the motor has to supply angular momentum ($M > 0$); when they move inward it must brake ($M < 0$).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^{P_i}$ twice reproduces $\mathbf{a}_1^{P_i} + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i}) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i}$ (Notes Eq. 3.27), and $dE_1/dt = 0$ along the equations of motion of part (b) (both checked symbolically with sympy, together with the equilibria of part (d)).
- **Dimensions.** $[k/d^2] = \mathrm{N \cdot m^2/m^2} = \mathrm{N}$; $[m\omega^2 x] = \mathrm{kg \cdot s^{-2} \cdot m} = \mathrm{N}$; $[M] = [m\omega x\dot{x}] = \mathrm{kg \cdot s^{-1} \cdot m \cdot m/s} = \mathrm{N \cdot m}$; $[\gamma] = \dfrac{\mathrm{m/s^2}}{\mathrm{s^{-2}\,m}} = 1$; $[\kappa] = \dfrac{\mathrm{N \cdot m^2}}{\mathrm{kg \cdot s^{-2} \cdot m^3}} = 1$.

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